

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2020

Mathematics

Paper 1

Ordinary Level

Time: 2 hours

300 marks

School Stamp

Running Total	
---------------	--

For Examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 14 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

--

Question 1 (Suggested maximum time: 10 minutes)

The first seven prime numbers are listed below.

2	3	5	7	11	13	17
---	---	---	---	----	----	----

(a) What is a prime number?

[illegible]

(b) Using the list of numbers above, write down

(i) three prime numbers whose **sum** is 25,

Answer = , and .

(ii) two prime numbers that have a **difference** of 9,

Answer = and .

(iii) three prime numbers whose **product** is 110.

Answer = , and .

[illegible]

(c) Write down the next two prime numbers after 17.

Answer = and .

(d) Katie says: “All prime numbers are odd”. Explain why Katie’s statement is **not** true.

[illegible]

Question 2

(Suggested maximum time: 10 minutes)

$$U = \{1, 2, 3, 4, 5\}$$

E is the set of even numbers between 1 and 5.

F is the set of factors of 4.

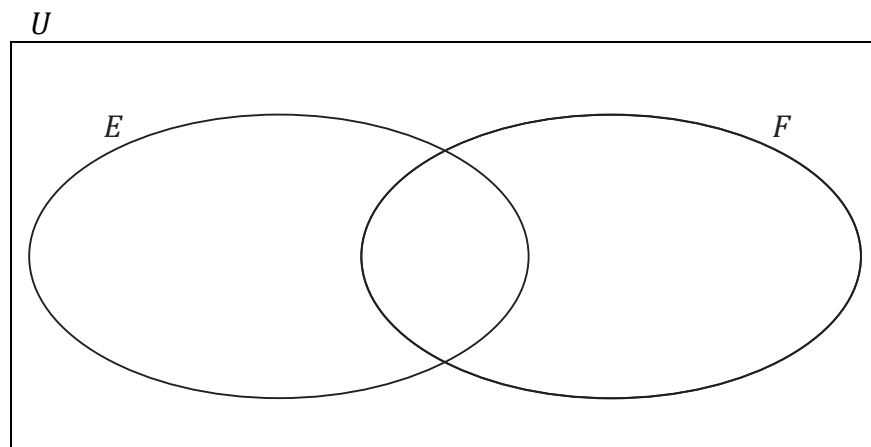
- (a) List the elements of the set E .

$$E = \{ \quad \quad \quad \}$$

- (b) List the elements of the set F .

$$F = \{ \quad \quad \quad \}$$

- (c) Use the information above to fill in the following Venn diagram.



- (d) List the elements of the set F' , where F' is the complement of the set F .

$$F' = \{ \quad \quad \quad \}$$

- (e) Put a tick (✓) in the correct box in each row of the table below, to show whether each statement is true or false.

Statement	Tick one only for each statement.	
	True	False
$E \cap F = F \cap E$		
$E \setminus F = \emptyset$		
$\#(E \cup F) = 3$		
$U = E \cup F$		

(Suggested maximum time: 10 minutes)

- | Day | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|------------------|--------|---------|-----------|----------|--------|----------|
| Temperature (°C) | −1 | 3 | 0 | 1 | −4 | −2 |

- | |
|--|
| |
|--|

-
- A diagram consisting of two identical rectangular boxes placed side-by-side. Each box contains a single horizontal line segment positioned in the lower half of the box. Between the two boxes is an equals sign (=).

- [illegible]

-
- A black and white line drawing of the front of a car. The car features a large, prominent oval grille with horizontal slats. On either side of the grille are rectangular headlights. Below the grille is a horizontal bumper bar. The car's hood and front fenders are also visible, along with side mirrors on the left and right. The drawing is a simple line art style, focusing on the car's silhouette and key features.

- [illegible]

[illegible]

- [illegible]

Question 4

(Suggested maximum time: 10 minutes)

Tickets for the local cinema cost €9 for an adult and €6 for a child.

The Murphy family of two adults and their children went to the cinema.

The total cost of the tickets was €42.

- (a)** How many children in the Murphy family went to the cinema?

- (b)** In total, 63 adults and 96 children attended the cinema at the same time as the Murphys. Work out the **total** cost of all the tickets that were sold.

[illegible]

- (c)** The film started at 13:50 and finished at 16:05. How **long** was the film?

[illegible]

- (d) Write the **ratio**, *the cost of an adult ticket : the cost of a child's ticket*, in its simplest form.

[illegible]

- (e) The Murphy family receive a voucher for 15% off their next visit to the cinema. Work out the total ticket cost for the next visit if they use the voucher and the same family members go the cinema.

[illegible]

(Suggested maximum time: 5 minutes)

(a) Bobbie got $\frac{3}{4}$ of the questions correct.

[illegible][illegible]

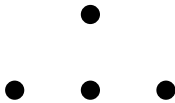
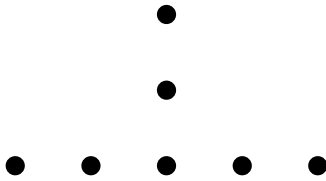
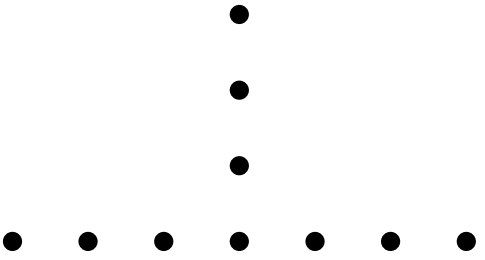
--

[illegible][illegible]

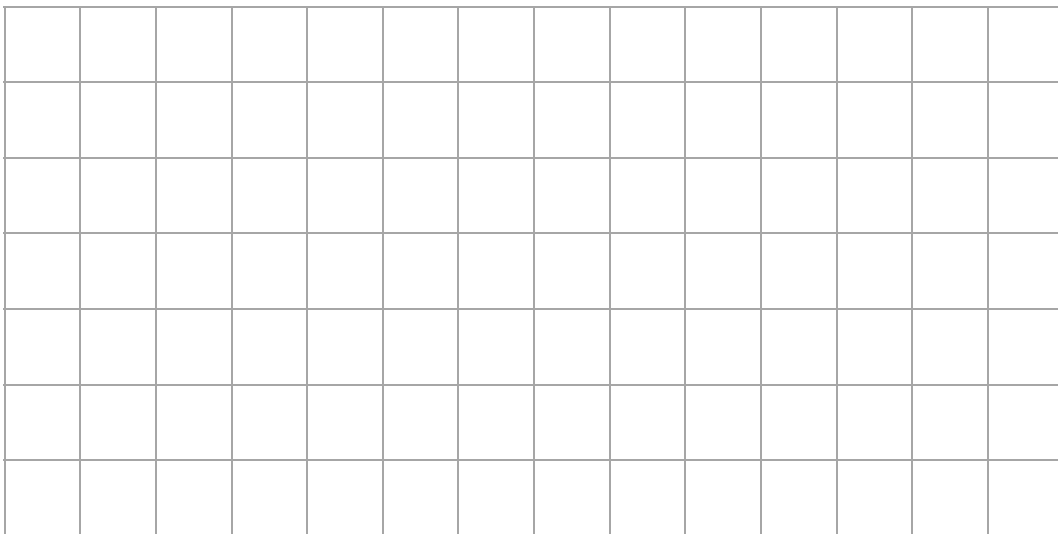
Question 6

(Suggested maximum time: 10 minutes)

The first patterns in a sequence of dots are shown.

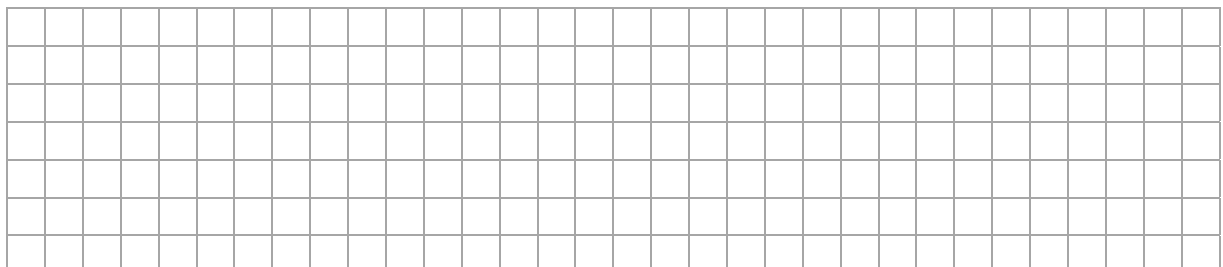
		
Pattern 1	Pattern 2	Pattern 3

(a) Draw Pattern 4 in the sequence.



(b) Fill in the table below to show the number of dots in each of the first six patterns.

Pattern	1	2	3	4	5	6
Number of dots						



- (c) What kind of sequence is made by the number of dots in each stage of the pattern?
Tick (✓) **one** box only. Give a reason for your answer.

linear

☐

quadratic

☐

exponential

☐

Reason:

- (d) The number of dots in Pattern n is:

$$3n + 1$$

Use this to work out what pattern has 2401 dots.

Question 7 (Suggested maximum time: 5 minutes)

Question 7 (Suggested maximum time: 5 minutes)

- (a)** Complete the table to show each number in the form 5^n , where $n \in \mathbb{N}$.

Number	Number in the form 5^n
25	
625	
$\frac{5^8}{5^4}$	
$5^3 \times 5^6$	
$(5^3)^2$	

- (b)** Write $10 \times 10 \times 10 \times 10 \times 10$ in the form 10^n , where $n \in \mathbb{N}$.

[illegible]

- (c) The distance between Dublin and New York is approximately 5 116 000 metres. Write this number in the form $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{N}$.

[illegible]

(Suggested maximum time: 10 minutes)

The graph shows a journey starting at 11:00. The distance from home increases from 0 km to 4 km between 11:00 and 11:20. It remains constant at 4 km from 11:20 to 11:40. The distance then decreases from 4 km back to 0 km between 11:40 and 12:00.

Time of day	Distance from home (km)
11:00	0
11:10	2
11:20	4
11:30	4
11:40	4
11:50	2
12:00	0

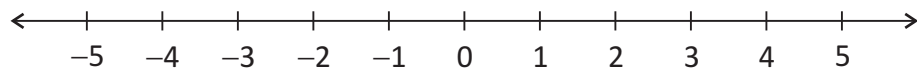
- [illegible]

Question 9

(Suggested maximum time: 5 minutes)

- (a) (i)** Solve the inequality $6x - 4 < 14, x \in \mathbb{N}$.

- (ii)** Graph your solution to $6x - 4 < 14$, $x \in \mathbb{N}$, on the number line given.



- (b) (i)** Factorise $x^2 + 7x + 12$.

[illegible]

- (ii)** Using your factors from (i) or otherwise, simplify $\frac{x^2 + 7x + 12}{x + 3}$.

[illegible]

- (iii)** Use your factors from (i) or otherwise to solve the equation $x^2 + 7x + 12 = 0$.

Question 10

(Suggested maximum time: 5 minutes)

- (a)** $\{(1, 3), (2, 6), (3, 9), (4, 12)\}$ are couples of a function f .

- (i) Write down the elements of the **domain** and the **range** of the function f .

Domain = $\{ \quad \quad \quad \}$

Range = { }

- (ii) Explain the relationship between the **domain** and the **range** of the function f .

[illegible]

- (b)** $f(x) = 6x - 4$.

- (i) Find the value of $f(4)$.

[illegible]

$f(4) =$	
----------	--

- (ii) Find the value of $f(-2)$.

[illegible]

$f(-2) =$	
-----------	--

Question 11

(Suggested maximum time: 5 minutes)

- (a)** Find the value of $3a + 7b - 8$, when $a = 5$ and $b = 2$.



- (b)** Multiply out and simplify $(3x + 1)(x - 2)$.

A large grid of 20 columns and 10 rows, intended for drawing. The grid is composed of thin black lines forming a series of small squares.

Question 12

(Suggested maximum time: 5 minutes)

- (a) Tom is T years old. Bill is 5 years older than Tom. Write Bill's age in terms of T .

[illegible]

- (b)** Tom says: “The sum of my age and Bill’s age can be represented by the expression $2T + 5$ ”. Show that Tom is correct.

[illegible]

- (c) Tom and Bill's ages add up to 21.
Write down an equation in T to represent this statement.

[illegible]

- (d)** Solve your equation to find Tom's age.

[illegible]

Question 13

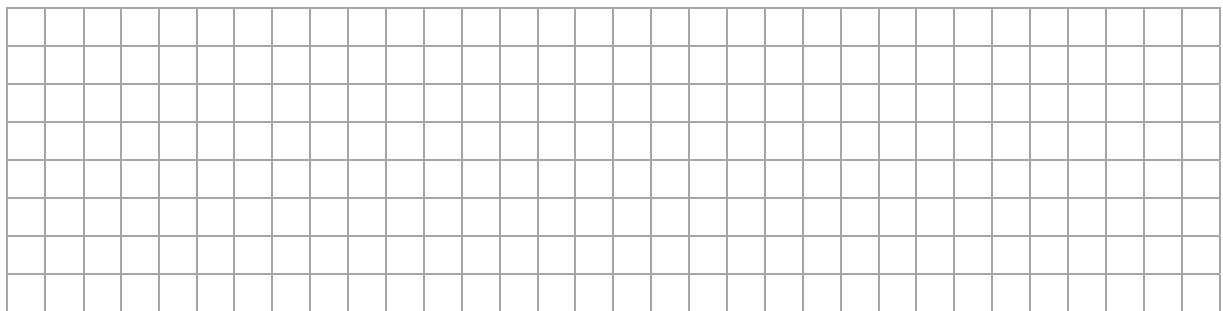
(Suggested maximum time: 15 minutes)

$g(x) = 2x + 4$ and $h(x) = 4 - x$ are two functions.

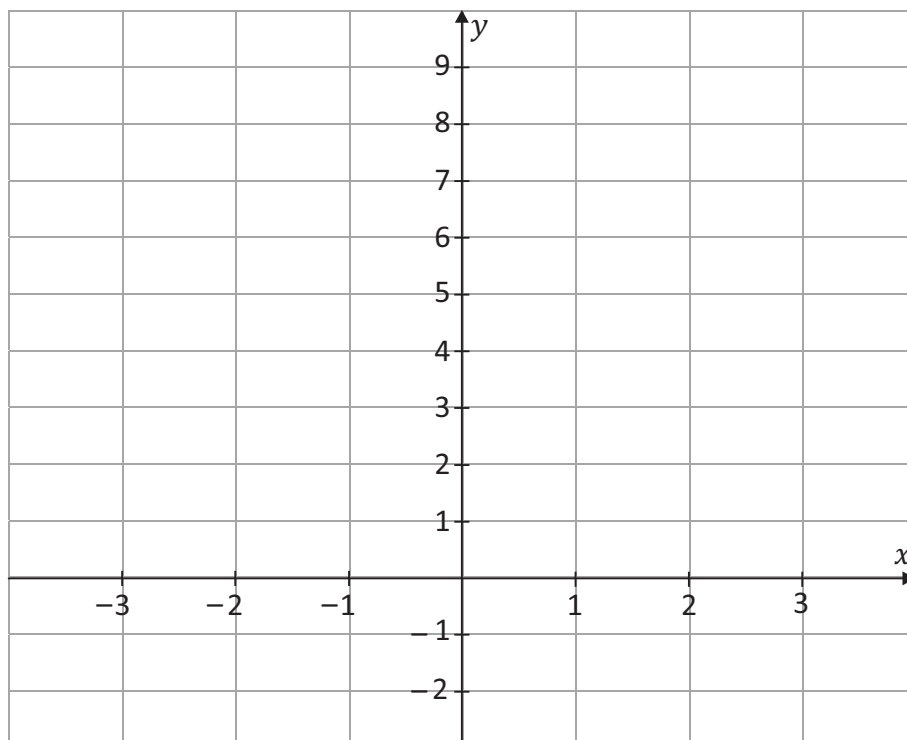
- (a) Complete the following tables for the functions $g(x)$ and $h(x)$ in the domain $-2 \leq x \leq 2$, for $x \in \mathbb{R}$.

x	$g(x)$
-2	
-1	
0	
1	
2	

x	$h(x)$
-2	
-1	
0	
1	
2	



- (b) Using your figures from the table, draw graphs of the functions $g(x) = 2x + 4$ and $h(x) = 4 - x$ on the diagram below in the domain $-2 \leq x \leq 2$, for $x \in \mathbb{R}$.



- (c) Using your graphs from part (b), write down the point of intersection of $g(x)$ and $h(x)$.

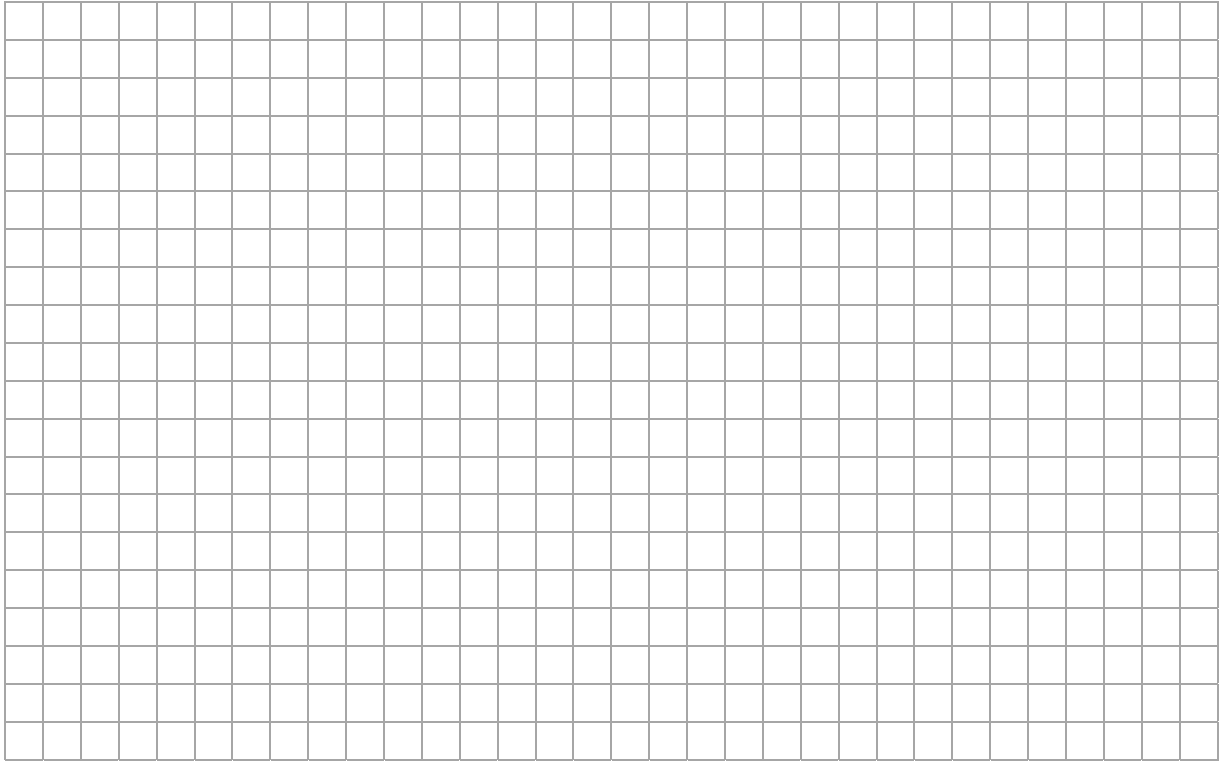
Answer =

(,)

- (d) Check your answer by using algebra to find the point of intersection of the lines.

$$y = 2x + 4$$

$$y = 4 - x$$



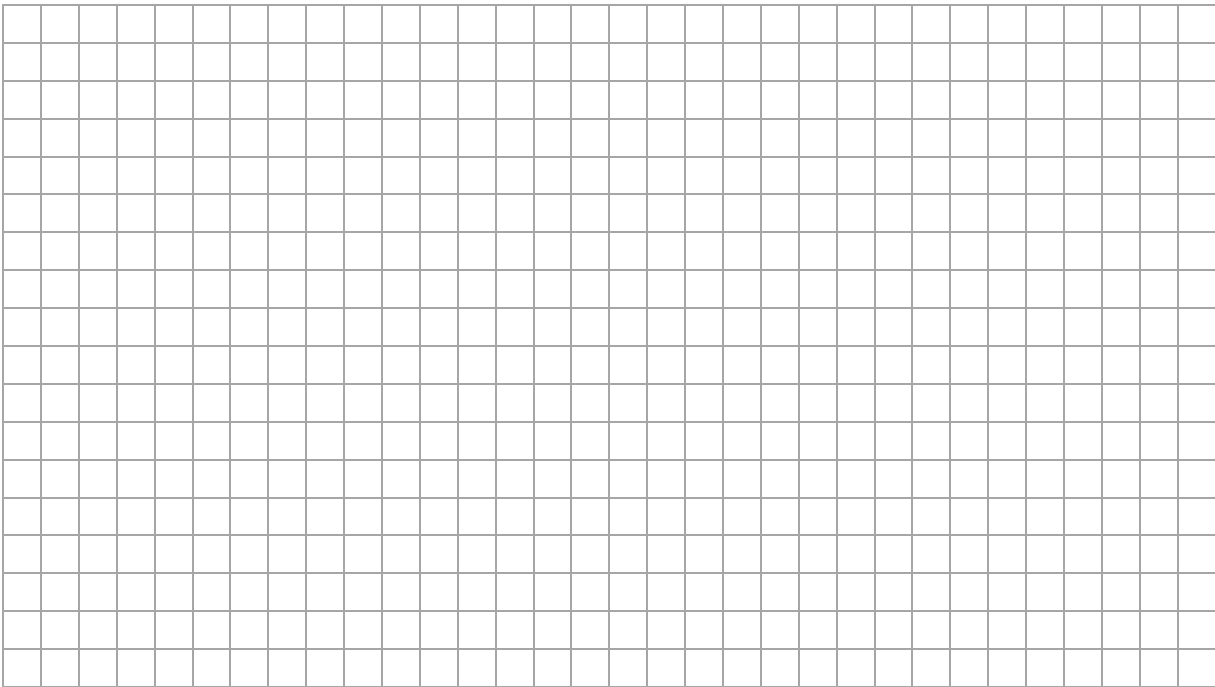
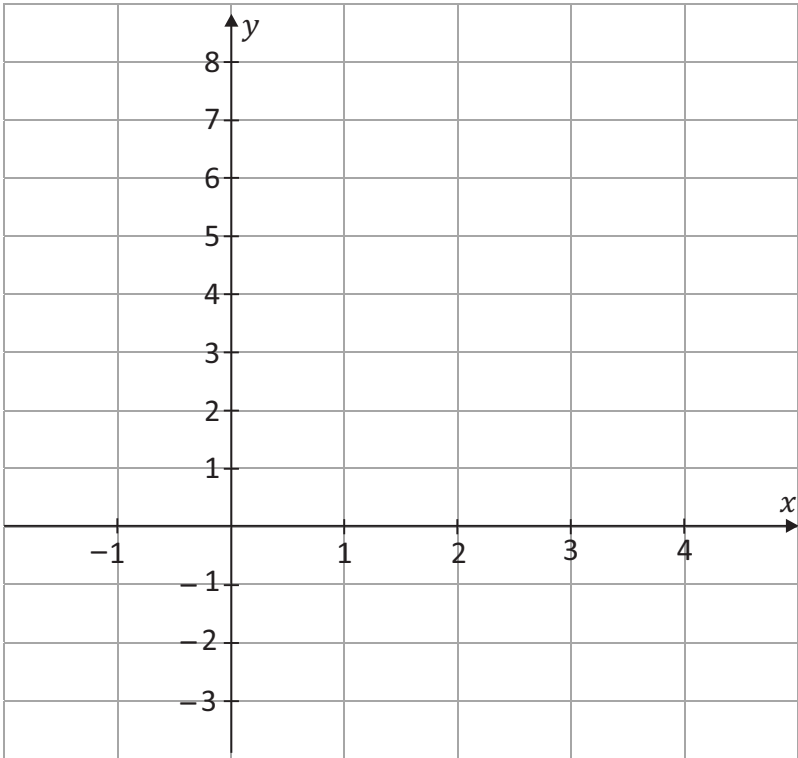
Question 14

(Suggested maximum time: 5 minutes)

Draw the graph of the function

$$y = x^2 - 4x + 2$$

on the axes below, for $-1 \leq x \leq 4$, $x \in \mathbb{R}$.



Label any extra work clearly with the question number and part.

This image shows a full page of blank graph paper. The grid consists of small, equal-sized squares formed by thin gray lines. There are no margins, text, or other markings on the page.

Page for extra work.

Label any extra work clearly with the question number and part.

This image shows a full page of blank graph paper. The grid consists of small, uniform squares formed by thin, light gray lines. There are no margins, text, or other markings on the page.

Pre-Junior Certificate Examination, 2020

Mathematics – Paper 1

Ordinary Level

Time: 2 hours

